

Exp No: 9

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

AIM:

To write an Embedded C program for interfacing Ultrasonic Sensor using Arduino Uno and Proteus.

SOFTWARE REQUIRED:

- Proteus 8 software.
- Arduino IDE software.

PROGRAM:

// Program Sketch

```
int trig = 10;
int echo = 9;
long duration;
int cm;
```

```
void setup() {
  // put your setup code here, to run once:
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}
```

```
void loop() { // Sketch to calculate ultrasonic distance
```

```
  digitalWrite(trig, LOW);
  delayMicroseconds(10);
```

```
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
```

```
  digitalWrite(trig, LOW);
  delayMicroseconds(10);
```

```
  duration = pulseIn(echo, HIGH);
```

```
  cm = (duration/2) * 0.034;
```

```
// Print Value on Serial Monitor
Serial.print(" Distance = ");
Serial.print(cm);
Serial.println(" cm");
}
```

OUTPUT :

